Theorem. *Let n be a positive integer. The probability that a uniform random function $f : [n] \to [n]$ has a unique cyclic vertex in its associated directed graph G_f is $1/n$.*

Proof. Let

$$\mathcal{F}_n = \{f : [n] \to [n]\}$$

and let

$$A_n = \{f \in \mathcal{F}_n : G_f \text{ has a unique cyclic vertex}\}.$$

The map $f \mapsto (f(1), \dots, f(n))$ is a bijection from $\mathcal{F}_n$ to $[n]^n$, so $|\mathcal{F}_n| = n^n$. By the uniform finite probability convention,

$$\Pr(A_n) = \frac{|A_n|}{|\mathcal{F}_n|} = \frac{|A_n|}{n^n}.$$

It remains to count $|A_n|$.

First record a finite-orbit fact. For $f : [n] \to [n]$, define $f^0(x) = x$ and $f^{k+1}(x) = f(f^k(x))$. For any $x \in [n]$, the $n+1$ vertices

$$f^0(x), f^1(x), \dots, f^n(x)$$

lie in the n-element set $[n]$, so two are equal. If $b > 0$ is the least index for which $f^a(x) = f^b(x)$ for some $a < b$, then

$$f^a(x), f^{a+1}(x), \dots, f^{b-1}(x)$$

are pairwise distinct and form a directed cycle in G_f. Hence every forward orbit eventually enters a directed cycle. If G_f has a unique cyclic vertex r, then the cycle through r has only the vertex r, so $f(r) = r$. Moreover, the eventual cycle reached by any vertex must be the loop at r; therefore every vertex reaches r after finitely many iterations of f.

[KEY STEP]. Fix $r \in [n]$, and set

$$\mathcal{A}_{n,r} = \{f : [n] \to [n] \colon r \text{ is the unique cyclic vertex of } G_f\}.$$

We prove that $|\mathcal{A}_{n,r}| = n^{n-2}$ for $n \geq 2$, and that $|\mathcal{A}_{1,1}| = 1$. If $n = 1$, the only function $[1] \to [1]$ has the single loop at 1, so $|\mathcal{A}_{1,1}| = 1$.

Assume $n \geq 2$. We construct a bijection

$$\Phi : \mathcal{A}_{n,r} \to [n]^{n-2}.$$

For $R \subseteq [n]$ containing r, call $a \in R \setminus \{r\}$ an R-leaf if no vertex $y \in R \setminus \{a\}$ has $f(y) = a$.

Given $f \in \mathcal{A}_{n,r}$, start with $R_1 = [n]$. For $k = 1, \dots, n-2$, choose a_k to be the least R_k-leaf in $R_k \setminus \{r\}$, set $w_k = f(a_k)$, and put $R_{k+1} = R_k \setminus \{a_k\}$. This is well-defined. Indeed, suppose at the beginning of a step that $r \in R_k$, that $f(R_k) \subseteq R_k$, and that every vertex of R_k reaches r. If no non-root R_k-leaf existed, then for every $u \in R_k \setminus \{r\}$ there would be a vertex $p(u) \in R_k \setminus \{u\}$ with $f(p(u)) = u$. Since $f(r) = r$, this $p(u)$ lies in $R_k \setminus \{r\}$. Iterating p in the finite set $R_k \setminus \{r\}$ would repeat and produce a directed cycle for f entirely outside r, contradicting the fact that every remaining vertex reaches r. Thus a least non-root R_k-leaf exists.

For such a leaf a_k, one has $f(a_k) \neq a_k$, since otherwise $a_k \neq r$ would be cyclic. Thus no vertex of R_k maps to a_k, and deleting a_k preserves $f(R_{k+1}) \subseteq R_{k+1}$. The remaining vertices still reach r, because a path from a remaining vertex to r cannot pass through a_k without some remaining vertex mapping to a_k. The induction continues until two vertices remain, $R_{n-1} = \{r, b\}$, and then closure, $f(r) = r$, and reachability force $f(b) = r$. Define

$$\Phi(f) = (w_1, \dots, w_{n-2}).$$

Conversely, given $w = (w_1, \dots, w_{n-2}) \in [n]^{n-2}$, start with $R_1 = [n]$. For $k = 1, \dots, n-2$, choose a_k to be the least element of $R_k \setminus \{r\}$ that does not occur among

$$w_k, w_{k+1}, \dots, w_{n-2}.$$

Such an element exists because $R_k \setminus \{r\}$ has $n-k$ elements and the suffix has length $n-k-1$. Define $f(a_k) = w_k$ and set $R_{k+1} = R_k \setminus \{a_k\}$. The assignment is legitimate: $w_k \neq a_k$, and w_k was not removed earlier, for if $w_k = a_j$ with $j < k$, then a_j would have appeared in the suffix $w_j, \dots, w_{n-2}$, contrary to the choice of a_j. After the $n-2$ steps, write the remaining set as $\{r, b\}$, and define $f(b) = r$ and $f(r) = r$. This defines a function $f : [n] \to [n]$. If a_k has index k, b has index $n-1$, and r has index n, then $f(v)$ has strictly larger index than v for every $v \neq r$. Hence every non-root vertex reaches r, while r is fixed, so the only directed cycle is the loop at r. Thus the decoding gives a map $\Psi : [n]^{n-2} \to \mathcal{A}_{n,r}$.

The maps Φ and Ψ are inverse maps. Let $f \in \mathcal{A}_{n,r}$ and let $\Phi(f) = w$. At encoding step k, for $y \in R_k \setminus \{r\}$,

$$y \text{ is an } R_k\text{-leaf} \quad \iff \quad y \text{ does not occur among } w_k, \dots, w_{n-2}.$$

If $y = w_j = f(a_j)$ for some $j \geq k$, then the still-present vertex $a_j \neq y$ maps to y, so y is not a leaf. Conversely, if y is not a leaf, choose $z \in R_k \setminus \{y\}$ with $f(z) = y$. Since $y \neq r$, the vertex z is neither r nor the final vertex b, because $f(r) = f(b) = r$. Thus $z = a_j$ for some $j \geq k$, and $w_j = y$. Therefore decoding $\Phi(f)$ chooses the same least vertex at every step and reconstructs f, so $\Psi(\Phi(f)) = f$.

Conversely, let $w \in [n]^{n-2}$ and let $\Psi(w) = f$. At decoding step k, the chosen a_k is an R_k-leaf, since it does not appear in the remaining suffix and neither b nor r maps to it. If $y < a_k$ is a non-root remaining vertex, then by the choice of a_k, the vertex y occurs in the remaining suffix, so some still-present vertex maps to y. Hence a_k is the least R_k-leaf, and the encoding records $f(a_k) = w_k$ at each step. Thus $\Phi(\Psi(w)) = w$.

Therefore Φ is a bijection from $\mathcal{A}_{n,r}$ to $[n]^{n-2}$. Since there are n^{n-2} words of length $n-2$ over $[n]$, we have

$$|\mathcal{A}_{n,r}| = n^{n-2} \qquad (n \geq 2).$$

Now for each $r \in [n]$, let

$$A_{n,r} = \{f \in \mathcal{F}_n : r \text{ is the unique cyclic vertex of } G_f\}.$$

Then

$$A_n = \bigsqcup_{r \in [n]} A_{n,r},$$

because every function with a unique cyclic vertex has exactly one such vertex $r \in [n]$, and a unique vertex cannot be two distinct elements. If $n = 1$, then $|A_n| = 1 = 1^{1-1}$. If $n \geq 2$, then each fixed-root class has size n^{n-2}, so

$$|A_n| = \sum_{r \in [n]} |A_{n,r}| = n \cdot n^{n-2} = n^{n-1}.$$

Thus $|A_n| = n^{n-1}$ for every positive integer n. Finally,

$$\Pr(A_n) = \frac{|A_n|}{n^n} = \frac{n^{n-1}}{n^n} = \frac{1}{n}.$$

This proves the theorem. $\square$